D222 with full NER tuning

CS781 Phase 2 legal named entity recognition | source-audited report

Task and evidence

Predict exact original-character entity spans in 12 legal categories. The source archive holds the actual code, configuration, run record, selected weights, and prediction ZIP. This report was produced after training from those records; it is not an executed historical notebook.

Architecture

D222 is a 24-layer, width-896 causal decoder with grouped-query attention: 14 query heads share two key/value heads. Every fourth layer is global; others use a 512-token local window. The NER head adds a character BiGRU, boundary logits, and CRF.

Path	Legal text -> Own byte BPE -> 24-layer D222 GQA -> Local/global attention
Output	Character BiGRU -> CRF + boundaries -> Span export

Training and validation

Method: Full fine-tuning. This export's archived held-out strict macro-F1 is 0.400950 on the 12-document local validation split. Its selected checkpoint and prediction ZIP hashes are in evidence/run_summary.json. The official test set has 35 document IDs; no hidden Codabench score is established by this archive.

Implementation map

Key original files: legal_d/model.py, legal_d/ner.py, legal_d/ner_train.py. The companion notebook explains the tensor route and reads archived evidence without presenting invented execution outputs.

Reproducibility boundary

The package includes exact source snapshots and small run records, plus the archived prediction ZIP. Large model weights, tokenizers, and course data stay in the separate complete archive. Original machine paths in config files need remapping for a new environment. The consensus rule was selected on validation; 100M/200M D222 exports were completed after the original deadline where marked.

Interpretation

Model variants change either pretrained backbone, parameter-update method, NER head, or pretrained checkpoint. A score difference does not isolate one factor unless all other choices match. The local held-out set is small, so its ranking should not be described as a hidden-test result.

Recorded model and adaptation

The source and run records together identify the following settings. Counts include the Phase 2 NER head. The method changes which backbone parameters are updated; it does not change the original Phase 1 tokenizer during this run.

Field	Archived value
Base family	D222
Adaptation	Full fine-tuning
Total parameters	222903806
Trainable parameters	222903806
Backbone LR	2e-05
Head LR	0.0005
Pretrained SHA-256	see run.json…
Selected NER SHA-256	773f9d8a7fb08f61e684…

Training signal and exact-span evaluation

For the common NER runner, the character tagger minimizes weighted cross-entropy over 25 BIO categories; A0 also matches its verified B1 teacher with temperature-2 KL. D222 adds CRF negative log-likelihood and a weighted boundary objective. HeaderCRF combines cross-entropy, CRF, and boundary losses. The validation scorer counts a prediction as correct only when the start, end, and label all match the gold span; macro-F1 averages the 12 per-label F1 values equally.

Observed optimization trace

These five entries are sampled from the archived training log; loss is a per-batch training value, not held-out macro-F1. Its decline alone cannot establish improved entity extraction.

Optimizer step	Epoch	Logged batch loss
1	1	3.2655
330	2	0.2960
670	5	0.3021
1000	6	0.2739
1330	8	0.2474

Label-level held-out scores

The following F1 values come from the archived selected validation prediction. Low-frequency labels are sensitive to a few documents, so read them beside their support and error counts in evidence/selection.json.

Entity label	F1	TP	FP	FN
APP	0.169	22	130	87
RESP	0.157	14	63	87
A.COUNSEL	0.310	13	26	32

Entity label	F1	TP	FP	FN
R.COUNSEL	0.308	12	35	19
JUDGE	0.523	17	13	18
WIT	0.000	0	25	21
AUTH	0.336	93	255	112
COURT	0.767	170	63	40
STAT	0.587	191	116	153
PREC	0.223	60	216	202
DATE	0.952	320	20	12
CASENO	0.479	69	81	69

Verification and reproducibility

The prediction ZIP has a single root prediction.json, passed CRC, and contains exactly 35 official test IDs. Its SHA-256 was compared with the completed run record. Each copied code file has a SHA-256 listed in evidence/SOURCE_HASHES.json. Exact re-execution requires the original selected weights, tokenizer and course data preserved in the complete archive; this compact package does not embed them or assert that historical notebook output cells existed.